

Seminar Class

HL

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Contents

1	Energy Estimates And Liouville Theorems For Harmonic Maps	5
1.1	Preliminaries	6
1.2	General lower energy estimates	6
1.3	Upper energy estimates for harmonic maps	8
1.4	Counterexamples and sharpness of the results	9
1.5	Geometric applications	12
2	A former report	15
2.1	Chen and Zhou's results	15
2.1.1	Main research subjects and notations	15
2.1.2	Foliated Riemannian manifolds	17
2.1.3	Kähler Foliations	18
2.2	Konderak and Wolak's results	19
2.2.1	Main research subjects and notations	19
2.2.2	Main results	21
2.2.3	Transversally harmonic maps and holonomy pseudogroups	21
2.2.4	The suspension construction	21

Chapter 1

Energy Estimates And Liouville Theorems For Harmonic Maps

Introduction

[RS00]

Theorem 1.0.1. *Let (M, g_o) be a complete m dimensional Riemannian manifold with a pole o , and, denoting by $r(x)$ the distance function from o , assume that the radial sectional curvature K_r satisfies*

$$-\frac{B^2}{(1+r^2)^{1+\epsilon}} \leq K_r \leq 0$$

for some $B > 0$ and $\epsilon > 0$. Let $u : (M, g_o) \rightarrow \mathbb{R}^n$ be a parallel mean curvature isometric immersion of (M, g_o) into $\mathbb{R}^n$. If there exists a fixed (oriented) m -plane P_o , a number α_o with

$$\alpha_o \geq \cos^l(\pi/(2\sqrt{kl})), \quad l = \min\{m, n-m\}, \quad k = \begin{cases} 1 & \text{if } l < 2 \\ 2 & \text{if } l \geq 2, \end{cases}$$

and a compact domain Ω such that for all tangential m -planes of $M \setminus \Omega$

$$\langle P, P_o \rangle \geq \alpha_o,$$

then $u(M)$ is an m -dimensional linear affine subspace of $\mathbb{R}^n$.

Theorem 1.0.2. *Let (M, g_o) be an m -dimensional ($m \geq 3$) Riemannian manifold with a pole o whose radial sectional curvature satisfies*

$$-B^2 \leq K_r \leq -A^2$$

with $(m-2)A \geq B \geq A > 0$. Let η be a non-increasing function such that

$$\eta(t) = o\left(e^{[(m-2)A-mB]t}\right) \quad \text{as } t \rightarrow +\infty.$$

If $u : (M, g_o) \rightarrow (N, h)$ is a harmonic map such that, for some q in N ,

$$\text{dist}_{(N, h)}(u(x), q) \leq \eta(r(x)) \quad \text{for } r(x) \gg 1,$$

then u is constant.

Proposition 1.0.1. *Let (M, g_o) be a manifold with a pole o . Denote by $r(x) = \text{dist}_{(M, g_o)}(x, o)$ and assume that the radial sectional curvature K_r satisfies Condition (0.3). Assume that $g = fg_o$ is a conformally related metric, with f such that*

$$\frac{\partial f}{\partial r} \geq -\frac{2}{m-2}[(m-2)A - B]f \quad \text{on } M \setminus \{o\}.$$

Let (N, h) be Riemannian manifold. Then any finite energy harmonic map

$$u : (M, g) \rightarrow (N, h)$$

is constant.

Proposition 1.0.2. *Let (M, g_o) be a complete, Ricci flat, m -dimensional ($m \geq 5$) manifold with a pole o , and let $r(x)$ be the distance function from o , and B_r the geodesic ball $\{r(x) < r\}$. Assume that the radial sectional curvature K_r satisfies*

$$-\frac{B^2}{1+r^2(x)} \leq K_r(x) \leq \frac{A^2}{1+r^2(x)} \quad \text{on } M,$$

for some $B \geq 0$ and $0 \leq A \leq 1/2$ such that

$$2\sqrt{1+4B^2} - (m-2)\sqrt{1-4A^2} < m-4.$$

If

$$\int_{B_r} \langle R, R \rangle dV_{g_o} = o\left(r^{\sqrt{1+4B^2} - (m-2)\sqrt{1-4A^2}/2}\right) \quad \text{as } r(x) \rightarrow +\infty$$

then (M, g_o) is flat.

1.1 Preliminaries

1.2 General lower energy estimates

Lemma 1.2.1. *Given a vector field X on M , and a V -valued p -form ω define the vector field W on M by the formula*

$$W = \frac{1}{2} \langle \omega, \omega \rangle X - p \sum_{k=1}^m \langle X \lrcorner \omega, e_k \lrcorner \omega \rangle e_k,$$

where $\{e_k\}$ is any local orthonormal frame on M . Then

$$\begin{aligned} \text{div}_g W &= \frac{1}{4} \langle \omega, \omega \rangle \sum_{r=1}^m (L_X g)(e_r, e_r) - \frac{p}{2} \sum_{s,t} \langle e_s \lrcorner \omega, e_t \lrcorner \omega \rangle L_X g(e_s, e_t) \\ &\quad + \langle \omega, X \lrcorner d\omega \rangle + p \langle X \lrcorner \omega, d^* \omega \rangle. \end{aligned}$$

Basic setting: Fix a complete background metric g_0 on M , and choose a point $o \in M$. Denote

$$\begin{aligned} r(x) &= \text{dist}(x, o); \\ B_R &= \{r < R\}. \end{aligned}$$

Now assume the metric g has the form

$$g = fg_0.$$

Then

$$\begin{cases} \nabla_g h = f^{-1} \nabla_{g_0} h \\ dV_g = f^{m/2} dV_{g_0} \\ dV_g|_{\partial B_R} = f^{(m-1)/2} dV_{g_0}|_{\partial B_R}. \end{cases}$$

(A1) *There exist smooth functions $\phi, \psi : [0, +\infty) \rightarrow [0, +\infty)$ such that $\phi(t), \psi(t) > 0$ for $t > 0$, $\phi'(0) = \psi'(0) = 1$, and $\phi^{(2k)}(0) = \psi^{(2k)}(0) = 0$ for every $k = 0, 1, \dots$, so that*

$$\frac{\phi'}{\phi} (g_0 - dr \otimes dr) \leq \text{Hess}_{g_0} r \leq \frac{\psi'}{\psi} (g_0 - dr \otimes dr). \quad (1.1)$$

Given $\gamma : [0, +\infty) \rightarrow [0, +\infty)$ with $\gamma^{(2k)}(0) = 0$, $k = 0, 1, 2, \dots$ and $\gamma(t > 0)$ for $t > 0$, let X be the gradient field:

$$X = \gamma(r(x)) \partial r = \nabla_{g_0} \Gamma, \quad (1.2)$$

where

$$\Gamma(r) = \int_0^r \gamma(t) dt.$$

Then **(A1)** is equivalent to say

$$\begin{aligned} & 2\gamma(r) \left\{ \frac{\phi'(r)}{\phi(r)} g_0 + \left[\frac{\gamma'(r)}{\gamma(r)} - \frac{\phi'(r)}{\phi(r)} \right] dr \otimes dr \right\} \\ & \leq L_X g_0 \\ & \leq 2\gamma(r) \left\{ \frac{\psi'(r)}{\psi(r)} g_0 + \left[\frac{\gamma'(r)}{\gamma(r)} - \frac{\psi'(r)}{\psi(r)} \right] dr \otimes dr \right\}, \end{aligned}$$

which implies, in terms of $g - fg_0$,

$$\begin{aligned} & \gamma(r) \left\{ \left(f^{-1} \frac{\partial f}{\partial r} + 2 \frac{\phi'(r)}{\phi(r)} \right) g + 2 \left(\frac{\gamma'(r)}{\gamma(r)} - \frac{\phi'(r)}{\phi(r)} \right) f dr \otimes dr \right\} \\ & \leq L_X g \\ & \leq \gamma(r) \left\{ \left(f^{-1} \frac{\partial f}{\partial r} + 2 \frac{\psi'(r)}{\psi(r)} \right) g + 2 \left(\frac{\gamma'(r)}{\gamma(r)} - \frac{\psi'(r)}{\psi(r)} \right) f dr \otimes dr \right\}. \end{aligned} \quad (1.3)$$

Lemma 1.2.2. *Assuming (A1), then within the cut locus of o , we have*

$$\begin{aligned} \operatorname{div} W \geq & \frac{1}{2} \langle \omega, \omega \rangle \gamma \left\{ \frac{(m-2p)}{2} f^{-1} \frac{\partial f}{\partial r} + (m-p) \frac{\phi'}{\phi} - p \frac{\psi'}{\psi} \right. \\ & \left. + \min \left(\frac{\psi'}{\psi} - \frac{\gamma'}{\gamma}, \frac{\gamma'}{\gamma} - \frac{\phi'}{\phi} \right) \right\}. \end{aligned}$$

Denote

$$\begin{aligned} v := & \left\{ \frac{(m-2p)}{2} f^{-1} \frac{\partial f}{\partial r} + (m-p) \frac{\phi'}{\phi} - p \frac{\psi'}{\psi} \right. \\ & \left. + \min \left(\frac{\psi'}{\psi} - \frac{\gamma'}{\gamma}, \frac{\gamma'}{\gamma} - \frac{\phi'}{\phi} \right) \right\}, \end{aligned} \tag{1.4}$$

i.e.,

$$\operatorname{div} W \geq \frac{1}{2} \langle \omega, \omega \rangle \gamma \cdot v.$$

(A2) If $v(x)$ is the function defined above, assume that $v(x)$ is nonnegative on $M \setminus \{o\}$ and not identically zero. Furthermore assume that there exists $R_0 > 0$ and a smooth function $F : [R_1, +\infty) \rightarrow (0, +\infty)$ satisfying F and $F' \geq 0$ and such that

$$v(x) \geq \frac{F'}{F}(r(x)) \quad \text{on } M \setminus B_{R_0}. \tag{1.5}$$

Theorem 1.2.1. *Assuming (A1) and (A2). Let ω be a harmonic V -valued p -form, then either ω vanishes identically, or $\exists R_1 > 0, C = C(\omega)$ s.t. for any $R > R_1$:*

$$\int_{B_R} \langle \omega, \omega \rangle dV_g \geq C \int_{R_1}^R F(t) \gamma^{-1}(t) dt. \tag{1.6}$$

1.3 Upper energy estimates for harmonic maps

Let $w : N \rightarrow \mathbb{R}$ be a convex function, and $u : M \rightarrow N$ is harmonic, then

$$\Delta_g w \circ u = \sum_i \operatorname{Hess} w(du(e_i), du(e_i)) \geq \sum_i \lambda \langle du(e_i), du(e_i) \rangle \geq 0,$$

where $\lambda(u(x)) \geq 0$ is the smallest eigenvalue of $\operatorname{Hess} w$ at $u(x)$. Recall that if w is strictly convex, then $\lambda(y) > 0$ for all $y \in N$.

Theorem 1.3.1. *Assume that there exist $R_0 \geq 0$ and a function α in $C^\infty(M \setminus B_{R_0})$ such that*

$$(i) \quad \Delta_g \alpha \leq 0 \quad \text{and} \quad (ii) \quad \alpha > 0 \quad \text{on } M \setminus B_{R_0}.$$

Suppose furthermore that there exists $A > 0$ such that

$$\int_{\partial B_r} |\nabla_g \alpha| dV_g|_{\partial B_r} \leq A \quad \forall r > R_0.$$

Let $u : (M, g) \rightarrow (N, h)$ be a smooth function which is harmonic on $M \setminus \bar{B}_{R_0}$ and such that $\overline{u(M \setminus \bar{B}_{R_0})}$ is compact in N and contained in a domain D supporting a strictly convex function w . Then, either

$$\alpha |du|^2 \in L^1(M, g)$$

or, having set

$$\mu(r) = \int_{\partial B_r} \alpha f^{-1/2} dV_g \Big|_{\partial B_r} \quad r \geq R_0,$$

we have

$$(i) \quad \mu^{-1}(r) \in L^1([R_0, +\infty))$$

and

$$(ii) \quad \int_{B_r} \alpha |du|^2 dV_g \leq C \left\{ \int_r^\infty \mu(t)^{-1} dt \right\}^{-1}$$

for some $C > 0$ and every $r \geq R_0$.

Taking $\alpha \equiv 1$, we get

Corollary 1.3.1. *Assume the same conditions in the Theorem 1.3.1. Define*

$$\mu_1(r) = \int_{\partial B_r} f^{-1/2} dV_g \Big|_{\partial B_r} \quad r \geq R_0,$$

and assume that

$$\mu_1^{-1} \in L^1([R_0, +\infty)).$$

Then the energy of u satisfies

$$\int_{B_r} |du|^2 dV_g = O \left(\left\{ \int_r^{+\infty} \mu_1(t)^{-1} dt \right\}^{-1} \right) \quad \text{as } r \rightarrow +\infty.$$

Corollary 1.3.2. *Assume the same conditions in the Theorem 1.3.1, also assume that μ^{-1} is not integrable at infinity so that $\alpha |du|^2$ is in $L^1(M, g)$, and that*

$$\iota_\alpha(r) = \inf_{\partial B_r} \alpha \rightarrow 0 \quad \text{as } r \rightarrow +\infty.$$

Then

$$\int_{B_r} |du|^2 dV_g = o(\iota_\alpha^{-1}(r)) \quad \text{as } r \rightarrow +\infty.$$

1.4 Counterexamples and sharpness of the results

Given smooth functions $\eta, \zeta : [0, +\infty) \rightarrow [0, +\infty)$, such that $\eta(t), \zeta(t) > 0$ for $t > 0$, and

$$\eta'(0) = \zeta'(0) = 1, \quad \text{and} \quad \eta^{(2k)}(0) = \zeta^{(2k)}(0) = 0 \quad \forall k = 0, 1, \dots,$$

we consider the metrics on $\mathbb{R}^m$ defined in polar coordinates by

$$ds^2 = dr^2 + \eta^2(r)d\theta^2 \quad \text{and} \quad g_o = dr^2 + \zeta^2(r)d\theta^2,$$

where $d\theta^2$ is the standard metric of unit sphere $\mathbb{S}^{m-1}$. Note the conditions above insure that η and ζ are smooth metric of $\mathbb{R}^m$, and denote the corresponding Riemannian manifolds by $M(\eta)$ and $M(\zeta)$, respectively. (For instance, $\sin(t), \sinh(t), t$, etc.)

Consider **rotationally symmetric maps** $F_\rho : M(\zeta) \rightarrow M(\eta)$ defined in polar coordinates by

$$F_\rho : (r, \theta) \mapsto (\rho(r), \theta) \quad \rho(0) = 0.$$

The energy of F_ρ on B_r is (up to a constant)

$$E_r(F_\rho) = \int_0^r \left[(\rho')^2 + \frac{m-1}{\zeta^2} \eta^2(\rho) \right] (s) \zeta^{m-1}(s) ds, \quad (1.7)$$

and therefore F_ρ is a (smooth) harmonic map if and only if ρ is a nonnegative solution of

$$\begin{cases} \rho''(r) + (m-1) \frac{\zeta'}{\zeta} \rho'(r) - \frac{m-1}{\zeta^2} \eta(\rho(r)) \eta'(\rho(r)) = 0 & r > 0, \\ \lim_{r \rightarrow 0^+} \rho(r) = 0. \end{cases} \quad (1.8)$$

Theorem 1.4.1. *Assume that there exist λ in $(0, 1)$, $a > 0$, $k > 1$, and $R_o > 0$ such that*

$$\begin{aligned} (i) \quad & (m-1) \lambda \frac{\zeta'}{\zeta}(r) \geq \frac{1}{r} + \frac{k}{r \log r} \\ (ii) \quad & (1-\lambda) \zeta(r) \zeta'(r) \geq ar(\log r)^k \quad \text{on} \quad [R_o, +\infty) \end{aligned} \quad (1.9)$$

Then there exists $L > 0$ such that, for every ρ_∞ in $(0, L]$, (1.8) has a solution ρ satisfying

$$\lim_{r \rightarrow +\infty} \rho(r) = \rho_\infty.$$

The proof used the following lemma.

Lemma 1.4.1 ([RR93]). *Suppose that there exist L_o in $(0, s_o]$ and a C^2 solution $\beta : (0, +\infty) \rightarrow (0, L_o)$ of the problem*

$$\begin{cases} \beta''(r) + (m-1) \frac{\zeta'}{\zeta} \beta'(s) - \frac{m-1}{\zeta^2} \eta(\beta(r)) \eta'(\beta(r)) \geq 0 \\ \lim_{r \rightarrow 0^+} \beta(r) = 0. \end{cases}$$

Then there exist $\rho_\infty < L_o$ and a smooth, strictly increasing solution ρ of (1.8) with the property that

$$\lim_{r \rightarrow +\infty} \rho(r) = \rho_\infty.$$

Remark 1.4.1. The conditions (1.9) can be satisfied if there exists $\delta > 1$ and $R_0 \gg 1$ s.t.

$$\zeta(t) = t(\log(t))^\delta + q(t), \quad \forall t \geq R_0 \quad (1.10)$$

where q is smooth satisfies $q'(t) = o((\log(t))^\delta)$, (i.e., $q(t) = o(t(\log(t))^\delta)$) as $t \rightarrow +\infty$. For instance, $q(t) = \sin(t)$, then K_r of $M(\zeta)$ does not have constant sign at infinity.

It follows that for every $R_1 < L$ there exists a harmonic map $F_\rho : M(\zeta) \rightarrow M(\eta)$ with the property that

$$F_\rho \left(M(\zeta) \setminus \overline{B(R_2)} \right) \subset B_{M(\eta)}(R_1)$$

for suitably large R_2 .

Example 1.4.2 (The lower bound in Theorem 1.2.1 sharp). Let $m \geq 3$, choose a smooth function satisfies (1.10). One can show that

$$\rho'(r) \asymp r^{-1}(\log r)^{-2\delta} \quad \text{as } r \rightarrow +\infty,$$

which implies

$$E_r(F_\rho) \asymp \int_1^r \zeta^{m-3}(s) ds \asymp r^{m-2}(\log r)^{(m-3)\delta} \quad \text{as } r \rightarrow +\infty.$$

On the other hand, since $M(\zeta)$ is a model, **(A1)** holds trivially with $\phi = \psi = \zeta$. Set

$$v(x) = (m-2) \frac{\zeta'}{\zeta}(r(x)),$$

then **(A2)** satisfies with $F(r) = \zeta^{m-2}(r)$. In this case, the conclusion of Theorem 1.2.1 says

$$E_r(F_\rho) \geq C \int_{R_0}^r \zeta^{m-3}(s) ds \asymp r^{m-2}(\log r)^{(m-3)\delta} \quad \text{as } r \rightarrow +\infty.$$

Example 1.4.3 (The conclusion of Corollary 1.3.2 is sharp). Let $m \geq 3$, and $\zeta(t) = t^\delta, \delta > 1$, for $t > R_0$. Then the validity of (1.9) can be checked. Thus, by Theorem 1.4.1, there exists a rotationally symmetric harmonic map F_ρ , and similar calculation shows

$$E_r(F_\rho) \asymp r^{\delta(m-3)+1} \quad \text{as } r \rightarrow +\infty.$$

On the other hand, if we choose $f \equiv 1, \alpha(r) = r^{1-(m-1)\delta}, r \geq R_0$, then Corollary 1.3.2 is applicable. One can check that μ^{-1} is not integrable at infinity, thus

$$E_r(F_\rho) = o(r^{\delta(m-1)-1}) \quad \text{as } r \rightarrow +\infty.$$

Note that $(m-1)\delta - 1 = (m-3)\delta + 1 + 2(\delta - 1)$, and let $\delta \rightarrow 1$, we find the conclusion of Corollary 1.3.2 cannot be improved in general.

Remark 1.4.2. There is not too much room to weaken Assumption (1.9). For instance, if $k = 0$, Theorem 1.4.1 is not true.

1.5 Geometric applications

To compare our work with that of Hildebrandt [Hil82], H. C. J. Sealey [Sea82], Z. Jin [Jin92], and D. Zhang [Zha94].

It's worth to observe that the choose

$$F(t) = \left\{ \tilde{f}^{(m-2p)/2}(t) \phi^{m-a}(t) \psi^{-b}(t) \gamma^d(t) \right\}^\sigma, \quad t \geq R_o, \quad (1.11)$$

is often appropriate.

Remark 1.5.1 (Compare with Hildebrandt and Sealey's results). Let $(\mathbb{R}^m, g_0)$ be Euclidean space and let o be the origin. Then v in (1.4) is

$$v(x) = (m-2) \left[\frac{1}{2} f^{-1} \frac{\partial f}{\partial r} + \frac{1}{r(x)} \right]. \quad (1.12)$$

Then $v \geq 0$ iff

$$r \frac{\partial f}{\partial r} \geq -2f \quad \text{on} \quad \mathbb{R}^m \setminus \{0\}. \quad (1.13)$$

Choose $a = b = 1, d = 0$ and $\tilde{f} \equiv 1$, in (1.11), then the condition in (A2) becomes

$$r \frac{\partial f}{\partial r} \geq 2(\sigma - 1)f \quad \text{on} \quad \mathbb{R}^m \setminus B_{R_0}. \quad (1.14)$$

Then Theorem 1.2.1 says

$$\int_{B_R} |du|^2 dV_g \geq \begin{cases} CR^{(m-2)\sigma} & \text{for } \sigma > 0 \\ C \log R & \text{for } \sigma = 0 \end{cases} \quad (1.15)$$

These estimates were obtained in [Hil82], ($\sigma > 0$) and [Sea82] ($\sigma = 0$), respectively.

By different choice, let $\tilde{f} = e^{2\lambda t}$, $\lambda > 0$ in (1.11), and assume

$$r \frac{\partial f}{\partial r} \geq 2[\sigma(\lambda r + 1) - 1]f \quad \text{on} \quad \mathbb{R}^m \setminus B_{R_0}, \quad (1.16)$$

one can get from Theorem 1.2.1 that

$$\int_{B_R} |du|^2 dV_g \geq CR^{\sigma(m-2)-1} e^{(m-2)\sigma\lambda R} \quad (1.17)$$

for large R . Note that (1.16) and (1.14) can be satisfied by choosing $f(x) = e^{2r(x)}$. Thus improving Hildebrandt's result.

Remark 1.5.2 (Compare with Jin and Zhang' results). Assuming (1.13), (1.14) and

$$f(x) \leq Cr^{2(\sigma-1)}(x) \quad \text{on} \quad \mathbb{R}^m \setminus B_{R_0},$$

one can show that if

$$u : (\mathbb{R}^m, g = fg_0) \rightarrow (N, h)$$

is a harmonic map such that, for a sufficiently large R_1 , $\overline{u(M \setminus \bar{B}_{R_1})}$ is compact in N and contained in a domain D supporting a strictly convex function, then u is a constant. Therefore obtained Theorem 1 of [Zha94], which in turn generalizes Theorem B in [Jin92].

To extend Zhang's results, in fact, one can prove the following.

Theorem 1.5.1. *Suppose the radial curvature of M satisfies the pinching condition:*

$$-\frac{B^2}{(1+r^2)^{1+\epsilon}} \leq K_r \leq 0,$$

and that

$$\begin{cases} r \frac{\partial f}{\partial r} \geq \frac{1}{m-2} [\sqrt{1+4B^2} + 3 - 2m] f & \text{on } B_{R_1} \setminus \{o\} \\ r \frac{\partial f}{\partial r} \geq 2(\sigma - 1)f & \text{on } M \setminus B_{R_1} \\ f \leq Cr^{2(\sigma-1)} & \text{on } M \setminus B_{R_1}. \end{cases}$$

then if $u : (\mathbb{R}^m, g = fg_0) \rightarrow (N, h)$ is a harmonic map such that $\overline{u(M \setminus \bar{B}_{R_1})}$ is compact in N and contained in a domain D supporting a strictly convex function, then u is a constant.

Chapter 2

A former report

2.1 Chen and Zhou's results

[1] Qun Chen and Wubin Zhou. Bochner-type formulas for transversally harmonic maps. International Journal of Mathematics, 23(03):1250003, March 2012

In this paper, Chen and Zhou establish some Bochner-type formulas for transversally harmonic maps and then give their applications, e.g., EellsSampson type theorem, by using transverse connection. They also research Kähler foliations and obtain a similar result to Siu [Siu80].

2.1.1 Main research subjects and notations

In [CZ12], Chen and Zhou investigated some properties of foliated maps between two **compact Riemannian foliations** $f : (M', g', \mathcal{F}') \rightarrow (M', g', \mathcal{F}')$. Consider the functional of f :

$$e_T(f) := \frac{1}{2} \|d_T f\|^2, \quad E_T(f) := \int_M e_T(f) d\text{vol}_M,$$

then f is call transversally harmonic in this paper if f is the extremal of $E_T(f)$ for any variation of f by **foliated** maps.

Remark 2.1.1. One result of the paper shows that this definition of transversally harmonic maps coincides with the one introduced by Konderak and Wolak [KW03] when the foliation $\mathcal{F}$ is harmonic, but not in general. In [KW03], f is called transversally harmonic iff the locally induced map $\bar{f}$ is harmonic in usual sense.

With the metric g , TM can be decomposed orthogonally into $T\mathcal{F} \oplus T\mathcal{F}^\perp$. Then there is a bundle map $\sigma : \nu(\mathcal{F}) \xrightarrow{\cong} T\mathcal{F}^\perp$ splitting the exact sequence

$$0 \rightarrow T\mathcal{F} \rightarrow TM \xrightarrow{\pi} \nu(\mathcal{F}) \rightarrow 0.$$

There is a unique **transverse connection** (or **Bott connection**?)

$$\nabla_X^T s = \begin{cases} \pi[X, \sigma(s)] & \text{for } X \in \Gamma(T\mathcal{F}) \\ \pi \nabla_X^M \sigma(s) & \text{for } X \in \Gamma(T\mathcal{F}^\perp). \end{cases} \quad (2.1)$$

The following formulas hold

$$\begin{aligned} X \langle s, t \rangle &= \langle \nabla_X^T s, t \rangle + \langle s, \nabla_X^T t \rangle, \\ \nabla_X^T \pi Y - \nabla_Y^T \pi X &= \pi[X, Y]. \end{aligned} \quad (2.2)$$

The corresponding **transverse curvature**

$$R^T(X, Y)s = \nabla_X^T \nabla_Y^T s - \nabla_Y^T \nabla_X^T s - \nabla_{[X, Y]}^T s,$$

and the **transversally sectional curvature**

$$K^T(U, W) = \langle R^T(U, W)W, U \rangle,$$

and the **transverse Ricci curvature**

$$\text{Ric}^T(U, W) = \langle R^T(U, e_\alpha)e_\alpha, W \rangle,$$

where $X, Y \in \Gamma(TM)$, $U, W \in \Gamma(T\mathcal{F}^\perp)$, and $\{e_\alpha, \alpha = n - q + 1, \dots, n\}$ is an orthogonal frame field of $T\mathcal{F}^\perp$.

$$K(X, Y) = K^T(X, Y) - 3\|\pi^\perp(\nabla_X^M Y)\|_g^2, \quad (2.3)$$

where $X, Y \in \mathcal{H}$.

Remark 2.1.2. The transverse curvature operator R^T has an important property that for any vector $V \in \mathcal{V}$, $V \lrcorner R^T = 0$.

Proposition 2.1.1 (E-L equation).

$$\tau_T(f) := \nabla_{e_\alpha} d_T f(e_\alpha) - d_T f(\nabla_{e_\alpha}^T e_\alpha) - d_T f(\kappa) = 0, \quad (2.4)$$

$e_i \in \mathcal{V}, e_\alpha \in \mathcal{H}$.

Remark 2.1.3. When $\kappa = 0$, then one can derive

$$\begin{aligned} \tau_T(f) &= (\nabla_{e_\alpha} d_T f)e_\alpha = \nabla_{df(e_\alpha)} d_T f(e_\alpha) - d_T f(\nabla_{e_\alpha}^T e_\alpha) \\ &= \nabla_{(df(e_\alpha))^\perp} d_T f(e_\alpha) - d_T f(\nabla_{e_\alpha}^M e_\alpha), \end{aligned} \quad (2.5)$$

which coincides with the transverse tension field introduced by Konderak and Wolak [KW03].

Definition 2.1.1 (Transverse second fundamental form).

$$B_T(f)(X, Y) := (\nabla_X d_T f)Y = \nabla_X d_T f(Y) - d_T f(\nabla_X^T Y) \in \Gamma(f^{-1}\nu(\mathcal{F}')), \quad (2.6)$$

$X, Y \in \mathcal{H}$. One has

$$B_T(f)(X, Y) = B_T(f)(Y, X),$$

and

$$\tau_T(f) = \text{tr } B_T(f) - d_T f(\kappa).$$

Remark 2.1.4. The map f between M and M' is said to be **horizontal** if df maps $T\mathcal{F}^\perp$ into $T\mathcal{F}'^\perp$.

Remark 2.1.5. Generally, a harmonic map is not necessarily transversally harmonic, and vice versa.

2.1.2 Foliated Riemannian manifolds

Theorem 2.1.2. *Let M and M' be two foliated Riemannian manifolds. Assume M is compact with nonnegative transverse Ricci curvature and the transversally sectional curvature of M' is nonpositive. If $f : M \rightarrow M'$ is transversally harmonic and the foliation $\mathcal{F}$ of M is harmonic, then $B_T(f) = 0$. In addition,*

1. *If the transverse Ricci curvature of M is positive somewhere, then f is transversally constant. That is to say, f maps the domain manifold M into some leaf of M' .*
2. *If the transversally sectional curvature of the target manifold M' is negative, then f is transversally constant or the rank of $d_T f$ is one everywhere.*

Proof. 1) **Bott connection and local transverse normal frame field.** For any $x \in M$, we can choose a local orthogonal frame field

$$\{e_A\}_{A=1,\dots,n} = \{e_i, e_\alpha\}_{i=1,\dots,p, \alpha=p+1,\dots,n}$$

of TM near a point x such that $e_\alpha \in A(\mathcal{F})$ and

$$\nabla_{e_\alpha}^T e_\beta|_x = \nabla_{e_i}^T e_\alpha|_x = 0,$$

where $e_i \in \Gamma(T\mathcal{F})$, $e_\alpha \in \Gamma(T\mathcal{F}^\perp)$, $p := n - q$.

2) **Bochner formula.** Compute $\nabla^2 d_T f$ and $\Delta_M e_T(f)$ and then take their integral. $\square$

Remark 2.1.6. $A(\mathcal{F})$ stands for the space of transverse infinitesimal automorphisms, i.e., those vector fields Y with $[Y, V] \in \mathcal{V}$ for any $V \in \mathcal{V}$.

Proposition 2.1.2 (2nd variation formula).

$$\left. \frac{d^2}{dt^2} E(f_t) \right|_{t=0} = - \int_M \langle \nabla^2 v - R^T(df(e_\alpha), v) d_T f(e_\alpha), v \rangle d\text{vol}_M, \quad (2.7)$$

where $\nabla^2 v$ is the trace Laplace operator on the induced bundle $f^{-1}\nu(\mathcal{F}')$.

Remark 2.1.7. If the target manifold M' has nonpositive transversally sectional curvature, then any transversally harmonic map into M' is stable.

2.1.3 Kähler Foliations

Definition 2.1.3. $\mathcal{F}$ is called **transversally holomorphic** if, in addition, there exists a holonomy invariant complex structure $J : \nu(\mathcal{F}) \rightarrow \nu(\mathcal{F})$ and $\mathcal{L}_V J = 0$ for any $V \in \mathcal{V}$.

Definition 2.1.4. $\mathcal{F}$ is called **Kähler** if it is transversally holomorphic and satisfies that $g_{\nu(\mathcal{F})}(Jr, Js) = g_{\nu(\mathcal{F})}(r, s)$ for $r, s \in \Gamma(\nu(\mathcal{F}))$, and $\nabla J = 0$, where ∇ is the unique transverse connection.

Definition 2.1.5. One can define that f is **transversally holomorphic** if $d_T f \circ J = J' \circ d_T f$, and f is transversally anti-holomorphic if $d_T f \circ J = -J' \circ d_T f$.

Remark 2.1.8. Obviously, f is transversally holomorphic (respectively anti-holomorphic) if and only if $\bar{\partial}_T f = 0$ (respectively $\partial_T f = 0$).

Remark 2.1.9. By the example of Heisenberg groups, it seems that transversally holomorphicity or transversally anti-holomorphicity may lead to some **monotonicity inequalities** for horizontal harmonic maps on CR-manifolds.

Definition 2.1.6. Let $f : M \rightarrow M'$ be a differential map from Sasaki manifold (M, ϕ, ξ, η, g) to Sasaki manifold $(M', \phi', \xi', \eta', g')$. We call f is **contact holomorphic** if

$$df \circ \phi(X) - \phi' \circ df(X) \text{ is collinear with } \xi',$$

and f is contact anti-holomorphic if

$$df \circ \phi(X) + \phi' \circ df(X) \text{ is collinear with } \xi'$$

for any $X \in \Gamma(TM)$. It is easy to see f is contact holomorphic if and only if f is transversally holomorphic. Hence, the contact holomorphicity of f implies its transverse harmonicity.

Theorem 2.1.7. *Let M is a $(p+2)$ -dimensional Riemannian manifold with a codimension 2-dimensional harmonic Kähler foliation, and the Ricci curvature Ric^M is bounded from below by $-a(1 + r^2 \ln^2(r+2))$, where r is the distance function from a fixed point $x_0 \in M$ and a is a positive constant. Suppose M' is a Kähler foliated manifold whose transversally holomorphic sectional curvature is a positive constant λ and $f : M \rightarrow M'$ is a transversally harmonic map, we have*

- (i) *If $e_T^-(f) \leq \frac{K}{\lambda} + \frac{1}{2}e_T^+(f) - \delta$, then f is transversally holomorphic;*
- (ii) *If $e_T^+(f) \leq \frac{K}{\lambda} + \frac{1}{2}e_T^-(f) - \delta$, then f is transversally anti-holomorphic, where δ is a positive constant, and K is the transversally holomorphic sectional curvature of M .*

Proof. Apply the **generalized maximum principle** for the function $(e_T^-(f) + C)^{-\frac{1}{2}}$. $\square$

Theorem 2.1.8. *Let f be a transversally harmonic map from compact Sasaki manifold M to compact Sasaki manifold M' , and M' has a strongly negative transverse curvature. If the rank of $d_T f$ is at least three at some points of M , then f is contact holomorphic (or contact anti-holomorphic).*

Proof. Adapted from [Siu80].

1) Define

$$\begin{aligned} \partial_T f : T\mathcal{F}^{\perp(1,0)} &\rightarrow T\mathcal{F}'^{\perp(1,0)}, & \bar{\partial}_T f : T\mathcal{F}^{\perp(0,1)} &\rightarrow T\mathcal{F}'^{\perp(0,1)}, \\ \partial_T \bar{f} : T\mathcal{F}^{\perp(1,0)} &\rightarrow T\mathcal{F}'^{\perp(0,1)}, & \bar{\partial}_T \bar{f} : T\mathcal{F}^{\perp(0,1)} &\rightarrow T\mathcal{F}'^{\perp(1,0)}. \end{aligned} \quad (2.8)$$

Compute a new Bochner formula for $\omega = \langle \partial_T \bar{f}(\varepsilon_i), \bar{\partial}_T f(\bar{\varepsilon}_j) \rangle \bar{\omega}^j \wedge \omega^i$:

$$\begin{aligned} \partial_B \bar{\partial}_B \omega &= (\langle \nabla_{\bar{\varepsilon}_k} \partial_T \bar{f}(\varepsilon_i), \nabla_{\varepsilon_l} \bar{\partial}_T f(\bar{\varepsilon}_j) \rangle \\ &\quad + \langle R^T(\partial_T f(\varepsilon_l), \bar{\partial}_T \bar{f}(\bar{\varepsilon}_k)) \partial_T \bar{f}(\varepsilon_i), \bar{\partial}_T f(\bar{\varepsilon}_j) \rangle) \omega^l \wedge \bar{\omega}^k \wedge \bar{\omega}^j \wedge \omega^i \end{aligned} \quad (2.9)$$

2) Establish unique continuation property of contact holomorphic maps.

3)...

□

2.2 Konderak and Wolak's results

[2] Jerzy J Konderak and Robert A Wolak. Transversally Harmonic Maps Between Manifolds with Riemannian Foliations. Quarterly Journal of Mathematics, 2003.

In this paper, f is called transversally harmonic iff the locally induced map $\bar{f}$ is harmonic in usual sense.

Konderak and Wolak give many examples of transversally harmonic maps. In particular, they apply the suspension method to obtain transversally harmonic maps, also examples of harmonic maps and harmonic morphisms.

Relates the harmonicity of an equivariant map, with respect to an action of Lie groups, to the harmonicity of the induced map between the spaces of orbits

2.2.1 Main research subjects and notations

In [KW03], Konderak and Wolak define a different notion of transversally harmonicity. They construct a global transversal tension field of a map f , then they define transversally harmonic maps as those maps for which this tension field vanishes.

Let $f : (M_1, \mathcal{F}_1) \rightarrow (M_2, \mathcal{F}_2)$ be a smooth foliated map between two Riemannian foliations, and $p_i : U_i \rightarrow \bar{U}_i$ be a Riemannian submersion that define the foliation $\mathcal{F}_i$. Then there is a map $\bar{f} : \bar{U}_1 \rightarrow \bar{U}_2$ such that the following diagram commutes:

$$\begin{array}{ccc} U_1 & \xrightarrow{f} & U_2 \\ p_1 \downarrow & & \downarrow p_2 \\ \bar{U}_1 & \xrightarrow{\bar{f}} & \bar{U}_2 \end{array}$$

Every notions considering on U_i can be reduced to on $\bar{U}_i$, which is the central idea of this article.

Remark 2.2.1. Let $\mathcal{F}$ be a foliation on a Riemannian n -manifold (M, g) . Then $\mathcal{F}$ is defined by a cocycle $\mathcal{U} = \{U_i, f_i, g_{ij}\}_{i \in I}$ modelled on a q -manifold N_0 , where

- $\{U_i\}_{i \in I}$ is an open covering of M ,
- $f_i : U_i \rightarrow N_0$ are submersions with connected fibres,
- $g_{ij} : N_0 \rightarrow N_0$ are local diffeomorphisms of N_0 such that $f_i = g_{ij}f_j$ on $U_i \cap U_j$.

The connected components of the trace of any leaf of $\mathcal{F}$ on U_i consist of fibres of f_i . The open subsets $N_i = f_i(U_i) \subset N_0$ form a q -manifold $N = \bigsqcup N_i$, which can be considered to be a transverse manifold of the foliation $\mathcal{F}$. The pseudogroup $\mathcal{H}_N$ of local diffeomorphisms of N generated by g_{ij} is called the *holonomy pseudogroup* of the foliated manifold $(M, \mathcal{F})$ defined by the cocycle $\mathcal{U}$.

Remark 2.2.2. In the language of cocycles, **Riemannian foliations** means that there exists a cocycle $\mathcal{U}$ such that the transverse manifold N is a Riemannian manifold, the transformations g_{ij} are local isometries, and the submersions f_i are Riemannian.

Remark 2.2.3. The leaf space can be a very complicated topological space, but we can look at a foliated manifold as a **desingularization** of its leaf space.

Definition 2.2.1. A *basic partial connection* on $(M, \mathcal{F}, g)$ is a sheaf operator D such that for each open subset U of M

$$D : \Gamma_b(U, T\mathcal{F}^\perp) \times \Gamma_b(U, T\mathcal{F}^\perp) \rightarrow \Gamma_b(U, T\mathcal{F}^\perp)$$

and for any $X, Y, Z \in \Gamma_b(U, T\mathcal{F}^\perp)$ and any $f, h \in C_b^\infty(U)$.

Example 2.2.2. Let ∇ be the Levi-Civita connection of g ; then for any open subset U of M and $X, Y \in \Gamma_b(U, \mathcal{F}^\perp)$ we define D as

$$D_X Y = (\nabla_X Y)^\perp.$$

For Riemannian foliations, let $\varphi : U \rightarrow \bar{U}$ be a Riemannian submersion defining the foliation $\mathcal{F}$ on an open set U . One has

$$d\varphi(D_X Y) = \nabla_{\bar{X}}^{\bar{g}} \bar{Y}$$

for X, Y basic.

There are given natural bundle maps

$$U_i : T\mathcal{F}_i^\perp \rightarrow TM_i, \quad \Pi_i : TM_i \rightarrow T\mathcal{F}_i^\perp.$$

Lemma 2.2.1. Let X be a local foliated section of $T\mathcal{F}_1^\perp \rightarrow M_1$; then $\Pi_2 df(X)$ is a foliated section of the bundle $f^{-1}T\mathcal{F}_2^\perp$.

The foliated sections of $f^{-1}T\mathcal{F}_2^\perp$ may also be described in the following commuted diagram:

$$\begin{array}{ccc} f^{-1}T\mathcal{F}_2^\perp|U_1 & \xrightarrow{d\varphi_2} & \bar{f}^{-1}T\bar{U}_2 \\ \uparrow s & & \uparrow \bar{s} \\ U_1 & \xrightarrow{\varphi_1} & \bar{U}_1 \end{array}$$

The partial connection D^2 on $T\mathcal{F}_2^\perp$ determines, in a unique way, a partial **pull-back connection** D^2 on $f^{-1}(T\mathcal{F}_2^\perp) \rightarrow M_1$, such that $D_X^2(Y \circ f) = (D_{df(X)^\perp}^2 Y) \circ f$ for X, Y basic.

Lemma 2.2.2. *For a section $s \in \Gamma_b(U_1, f^{-1}(T\mathcal{F}_2^\perp))$, one has*

$$d\varphi_2(D_X^2 s) = \bar{\nabla}_X^2 \bar{s}.$$

Lemma 2.2.3. *For Z_1, Z_2 foliated vector fields,*

$$d\varphi_2(D(\Pi_2 df U_1)(Z_1, Z_2)) = (\bar{\nabla} df)(\bar{Z}_1, \bar{Z}_2)$$

Definition 2.2.3. The transversal tension field of f is defined by

$$\tau_b(f)_x = \text{trace}_{T\mathcal{F}_1^\perp} D(\Pi_2 df U_1) = \sum_{\alpha=1}^{q_1} D(\Pi_2 df U_1)(X_\alpha, X_\alpha),$$

where $X_1, \dots, X_{q_1}$ is an orthonormal basis of $T_x\mathcal{F}_1^\perp$.

2.2.2 Main results

Theorem 2.2.4. *The map f is transversally harmonic if and only if f is harmonic.*

Remark 2.2.4. There is no if and only if relationship between the harmonicity and the transversal harmonicity of f .

Example 2.2.5. Product manifolds, warped manifolds, ...

Theorem 2.2.6. *Let $f : (M_1, \mathcal{F}_1) \rightarrow (M_2, \mathcal{F}_2)$ be a leaf preserving harmonic map between two manifolds with Riemannian foliations. Moreover, assume that all leaves of the minimal foliation $\mathcal{F}_1$ are minimal, the foliation $\mathcal{F}_2$ is totally geodesic, and f is horizontal, that is, for all $df(T\mathcal{F}_1^\perp) \subset T\mathcal{F}_2^\perp$, the map f is transversally harmonic.*

2.2.3 Transversally harmonic maps and holonomy pseudogroups

Theorem 2.2.7. *Let $(M_1, \mathcal{F}_1), (M_2, \mathcal{F}_2), (\tilde{M}_1, \tilde{\mathcal{F}}_1)$ and $(\tilde{M}_2, \tilde{\mathcal{F}}_2)$ be four foliated manifolds by Riemannian foliations. Let $f : (M_1, \mathcal{F}_1) \rightarrow (M_2, \mathcal{F}_2)$ and $\tilde{f} : (\tilde{M}_1, \tilde{\mathcal{F}}_1) \rightarrow (\tilde{M}_2, \tilde{\mathcal{F}}_2)$ be two foliated maps. Let us also assume that the holonomy pseudogroups of $(M_1, \mathcal{F}_1)$ and $(\tilde{M}_1, \tilde{\mathcal{F}}_1)$ are equivalent as well as those of $(M_2, \mathcal{F}_2)$ and $(\tilde{M}_2, \tilde{\mathcal{F}}_2)$. If $\tilde{\mathcal{K}}(f) = \mathcal{K}(\tilde{f})$, then the map f is transversally harmonic if and only if the map $\tilde{f}$ is.*

2.2.4 The suspension construction

Use the suspension construction for constructing foliation preserving maps between foliated manifolds, and present examples of transversally harmonic maps between foliated manifolds obtained from Hopf fibrations (considered as foliated manifolds).

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